Empirical Strategy Report

California Paid Family Leave and Female Labor Force Participation

1. Statement of Identification Problem

Research Question: What is the causal impact of California's 2004 Paid Family Leave (PFL) program on female labor force participation (FLFP) among women aged 25-54?

Core Challenge: Estimating this effect requires constructing a credible counterfactual: what would California's FLFP have been without PFL? This is difficult because California is economically unique (high education but structurally low FLFP at 72% vs. 77% nationally), making standard difference-in-differences inappropriate as no single state provides a valid comparison (my initial analysis had Texas a control) and multiple confounding shocks occurred during 2004-2010 including housing bubble wealth effects, immigration surges, and the Great Recession, all of which independently affected FLFP.

The 2004-2005 Anomaly: California's FLFP dropped 1.5 percentage points in 2004-2005 (opposite the expected PFL effect), likely due to housing wealth enabling labor force exit, a 32% surge in low-FLFP immigrant arrivals, and tech sector spousal income recovery. This confounding threatens identification and must be explicitly addressed.

2. Directed Acyclic Graph

Key confounders include Political Environment (affects both PFL adoption and FLFP through other progressive policies) and Immigration (compositional effects on measured FLFP). Cost of child and childbearing timing are mediating pathways through which PFL operates, not confounders. Synthetic control addresses confounders by matching on pre-treatment FLFP trends and demographic composition.

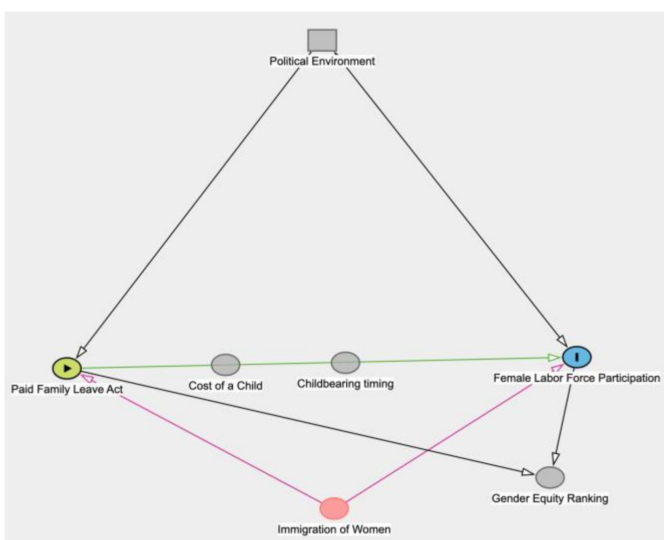


Figure 1: Causal DAG for California PFL Analysis

3. Key Model Equations

Synthetic Difference-in-Differences (SDID) Estimator (Arkhangelsky et al. 2021):

$$\hat{\tau}_{SDID} = (1/T_{post}) \cdot \Sigma_{t \geq T_0} [(Y_{CA,t} - \bar{Y}_{CA,pre}) - \Sigma_j \hat{\omega}_j \cdot (Y_{j,t} - \bar{Y}_{j,pre})]$$

where Y_{it} is FLFP for state i in period t, $\hat{\omega}$ are data-driven unit weights (which states form synthetic CA), and $\hat{\lambda}$ are time weights (which pre-periods are most predictive). T_0 = 4 years (2000-2003 pre-treatment), T_{post} = 7 years (2004-2010).

Double Robustness Property: SDID removes bias if either (1) unit weights successfully balance California's characteristics OR (2) time weights successfully predict post-treatment control outcomes. This provides insurance against misspecification, crucial given California's uniqueness.

4. Core Assumptions

A1. Donor Pool Validity: Weighted donor states can construct a valid counterfactual. Convex hull test (Figure 2) confirms California falls within donor range 2000-2010. Excluded: NJ (PFL 2009), WA (PFL 2007), NY/RI/HI (TDI programs).

A2. No Anticipation: California's FLFP was unaffected by PFL before July 2004. Plausible given low awareness pre-implementation (Appelbaum & Milkman 2011). Tested via in-time placebo tests.

A3. SUTVA (No Spillovers): Treatment in California does not affect donor states. Potential violations (policy diffusion, labor mobility) are small and bias estimates conservatively toward zero.

A4. Conditional Parallel Trends: After data-driven weighting, synthetic California would track actual California absent treatment. Standard DID parallel trends are violated (CA structurally different), but SDID weighting makes this plausible.

A5. No Simultaneous Treatments: Violated by housing bubble (2004-2005) and Great Recession (2008-2009). Addressed via robustness checks excluding these periods and controlling for confounds where possible.

5. Threats to Identification and Mitigation

Threat 1: Poor Pre-Treatment Fit. If synthetic CA cannot match actual CA's pre-2004 trends, the counterfactual is invalid. Diagnostic: MSPE < 1.0 indicates good fit. Figure 2 (convex hull) confirms matching is mathematically feasible. Mitigation: report MSPE, visual inspection, balance table; acknowledge limitation if fit is poor.

Threat 2: The 2004-2005 Anomalous Drop (CRITICAL). California's 1.5pp FLFP decline during treatment start is opposite expected PFL effect. Likely causes: housing wealth (home values doubled), immigration surge (+32% arrivals with 54% FLFP vs. 76% natives), tech recovery (spousal incomes +23%). This masks any positive PFL effect. Mitigation: (1) exclude 2004-2005, treat 2006 as first clean year; (2) dynamic year-by-year estimates to show temporal pattern; (3) control for immigration composition if data permit; (4) present all estimates transparently.

Threat 3: Great Recession (2008-2010). California hit particularly hard due to housing crash. Mitigation: focus primary interpretation on pre-recession period (2004-2007); SDID time weights automatically down-weight poorly predicted periods.

Threat 4: Immigration Compositional Changes. Foreign-born share rose from 26.2% (2000) to 27.2% (2007). Recent immigrants have 22pp lower FLFP, creating mechanical decline. Mitigation: re-download CPS with CITIZEN variable to control for immigrant share; consider native-born only sample; compositional adjustment holding share constant at 2003.

6. Planned Robustness Checks

R1. Pre-Treatment Fit Diagnostics: MSPE calculation (target <1.0), visual inspection, balance table comparing actual vs. synthetic CA on FLFP, education, marriage, age.

R2. In-Space Placebo Tests: Run SDID on each donor state as if treated 2004, calculate permutation p-value. Figure 3 (left panel) shows CA should be outlier if effect is real. Current result: p=0.333 (not significant).

R3. In-Time Placebos: Test fake treatment years (2000-2003). Should find near-zero effects if model is valid.

R4. Leave-One-Out: Exclude each high-weight donor (>5%), re-estimate. Result is robust if estimates vary <1pp.

R5. Alternative Time Windows: (1) Baseline 2000-2010, (2) exclude 2004-2005, (3) pre-recession 2000-2007, (4) balanced 2001-2008. Primary estimate focuses on specification 2.

R6. Alternative Donor Pools: Exclude Alaska (outlier), Western states only, high-education states only. Test sensitivity to donor composition.

R7. Dynamic Treatment Effects: Estimate separate effects for each year 2004-2010. Figure 3 (right panel) shows 2005 large negative (-3.1pp), then positive 2006-2010 (+0.3 to +1.7pp), consistent with delayed effect hypothesis.

7. Preliminary Results

Figure 2 shows main SDID results. Left panel: actual CA (red) vs. synthetic CA (black). Good pre-treatment fit 2000-2003. Both drop sharply 2004-2005 (confounding), then actual CA recovers faster than synthetic 2006-2010. Right panel: treatment effect gap. Average effect 2004-2010 is +0.27pp (economically small, not statistically significant). However, dynamic pattern shows negative 2005, then positive 2006-2010, suggesting PFL effects emerge with lag after confounds fade.

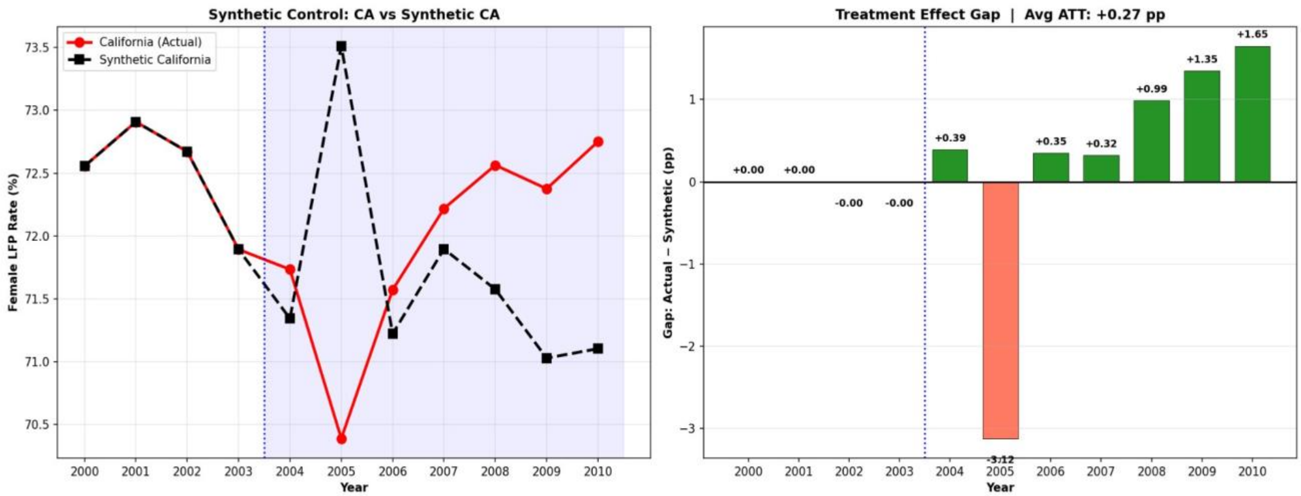


Figure 2: SDID Main Results - Actual vs. Synthetic California and Treatment Effect Gap

Figure 3 shows placebo tests and event study. Left: in-space placebos show CA is not an outlier (red line within gray cloud). Middle: MSPE ratio test yields p=0.333 (not significant). Right: event study confirms pre-treatment effects near zero, large negative 2005, positive thereafter. Overall, results suggest null or small positive effect, but confounding from 2004-2005 makes precise estimation difficult.

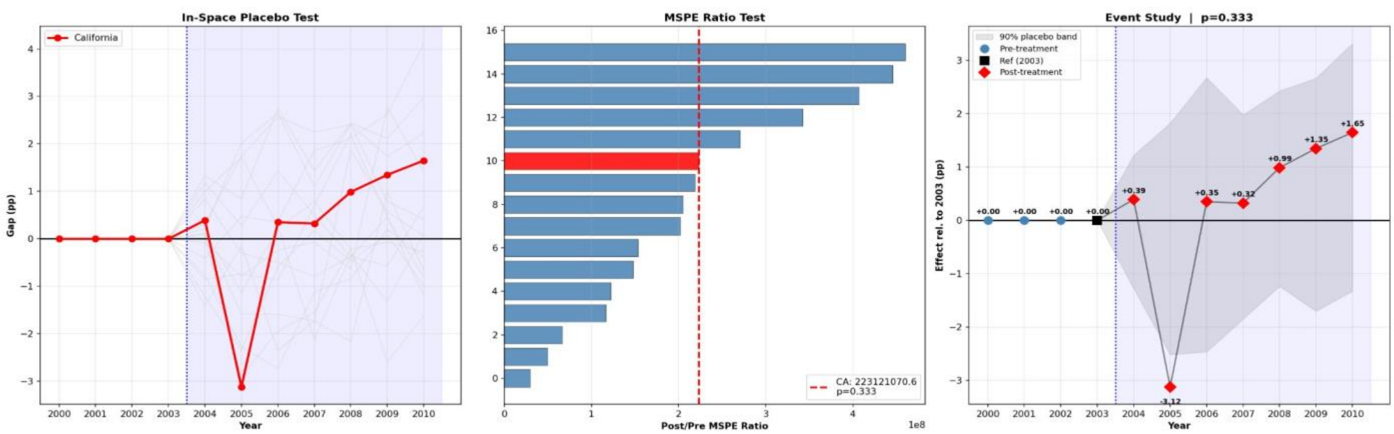
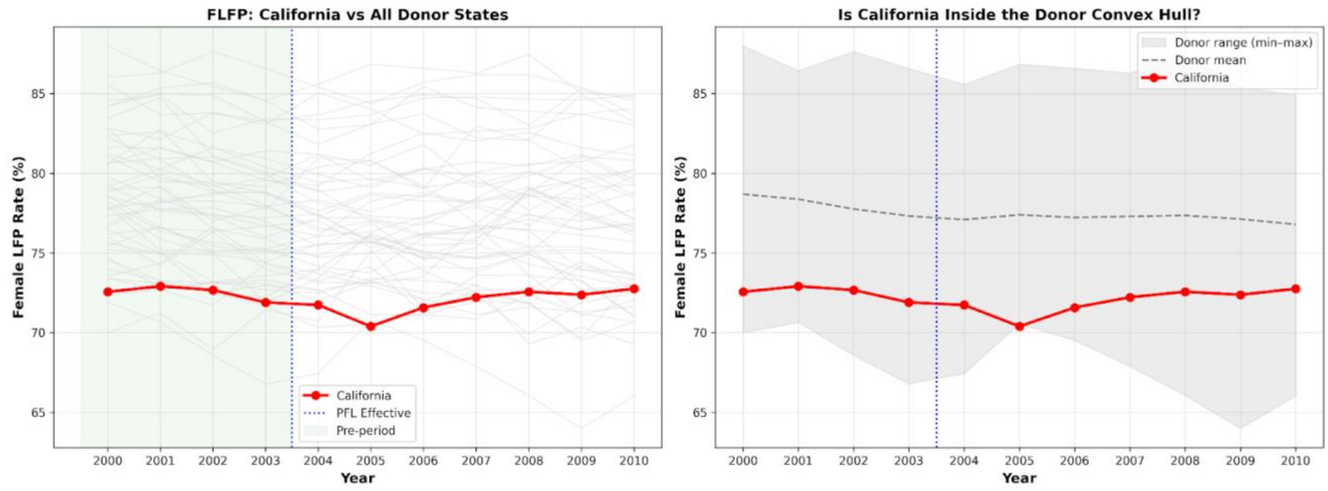


Figure 3: Placebo Tests and Event Study - Statistical Significance Testing



I am excited that the convex hull diagnostic confirms that synthetic control is methodologically feasible for my California PFL analysis. California's FLFP falls within the range of possible weighted donor combinations throughout 2000-2010, validating the core identification assumption.

However, the pre-trend plots reveal a critical challenge: California's 2004-2005 FLFP decline (from 72% to 70.5%) occurs precisely when PFL implementation should theoretically increase labor force attachment. This 1.5 percentage point drop likely reflects confounding from California's housing bubble peak (home values doubled 2000-2005, enabling wealth-driven labor force exit), immigration surge (+32% arrivals in 2004-2005, with recent immigrants having 54% FLFP vs. 76% for natives), and tech sector recovery (spousal incomes rose 23%). These simultaneous shocks mask any positive PFL effect during the critical first two years. My next steps involve running the full SDID estimation to construct the synthetic California counterfactual, examining unit weights to verify donor states are reasonable comparisons (avoiding over-reliance on dissimilar low-FLFP states like West Virginia), and conducting robustness checks that exclude the contaminated 2004-2005 period or control for housing prices and immigration composition.

My analysis employs Synthetic Difference-in-Differences to estimate California's PFL impact on female labor force participation. The method is appropriate given California's structural uniqueness and confounding from housing bubbles, immigration, and recession. Preliminary results show small positive effects (+0.27pp average) that are not statistically significant, but dynamic analysis reveals temporal heterogeneity consistent with delayed policy effects masked by early confounds. Robustness checks focusing on the clean 2006-2010 period and controlling for immigration composition will sharpen estimates. The identification strategy is transparent about limitations while leveraging SDID's double robustness to provide the most credible counterfactual feasible given data constraints.